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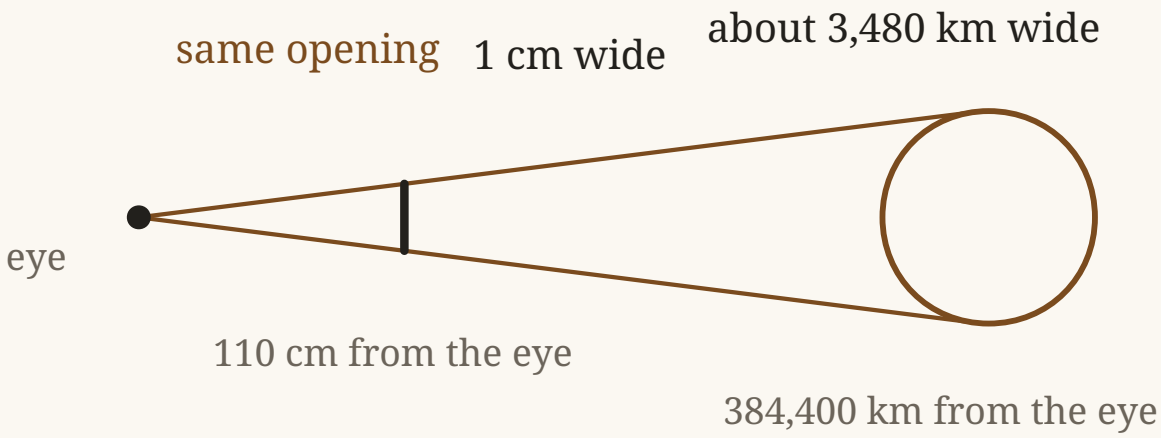
Thumb to Moon: how wide it looks is not how wide it is

A thumb held up at the Moon shows how wide the Moon looks. It does not show how many kilometers across the Moon is.

Call that look the opening: the small wedge of view the Moon fills. A small thing near your eye can fill the same opening as a large thing far away. The look matches. The sizes do not.



The same opening. A 1 cm mark at 110 cm, and the Moon about 3,480 km wide at 384,400 km. The stretch is shortened.

The drawing pulls the Moon in so both fit. They share one opening. Read the sizes from the labels, not from the lengths on the page.

A thumb is usually too wide to be the Moon's match. Picture a thumb about 2 centimeters across, about 60 centimeters from the eye. Yours may differ. This is a picture, not a ruler. In that picture the Moon fills a bit more than a quarter of the thumb, not the whole thumb. Do not look at the Sun this way.

The real width needs the distance, and the distance has to come from somewhere else. A thumb does not measure it. A photo of the Moon does not measure it. The photo only keeps the opening.

NASA's Moon facts give the average distance as 384,400 kilometers. The same page gives a radius of about 1,740 kilometers. The radius is the line from the center to the edge. The diameter is the width straight through the middle, two times the radius, so about 3,480 kilometers.

Check the round picture. Multiply 110 by 3,480.

100 times 3,480 is 348,000. 10 times 3,480 is 34,800. Added, that is 382,800.

NASA's distance is 384,400. The miss is 1,600 kilometers, small beside 384,400. So the Moon is about 110 times farther away than it is wide.

A mark 1 centimeter wide, held 110 centimeters from the eye, is also about 110 times farther away than it is wide. It matches the Moon's look. The mark is narrower than a thumb, and 110 centimeters is longer than a comfortable arm. That is why the thumb covers more.

An older map, not the method

A separate sheet draws a circle about 2,100 miles across on a map of the United States. It is a picture of a rounded width. It is not how the width is found. The Moon's width on a US map (PDF).

Sources

- NASA Moon facts: average distance 384,400 kilometers, radius about 1,740 kilometers.

Key facts

- A thumb shows how wide the Moon looks. It does not give the width in kilometers.
- The same opening can be a small near thing or a large far thing. A thumb is usually wider than the Moon's look.
- NASA's average distance is 384,400 kilometers. The width is about 3,480 kilometers, twice a radius of about 1,740.
- 110 times 3,480 is 382,800, close to 384,400. A 1 centimeter mark at 110 centimeters matches that picture.

The printed page is Thumb to Moon (PDF).